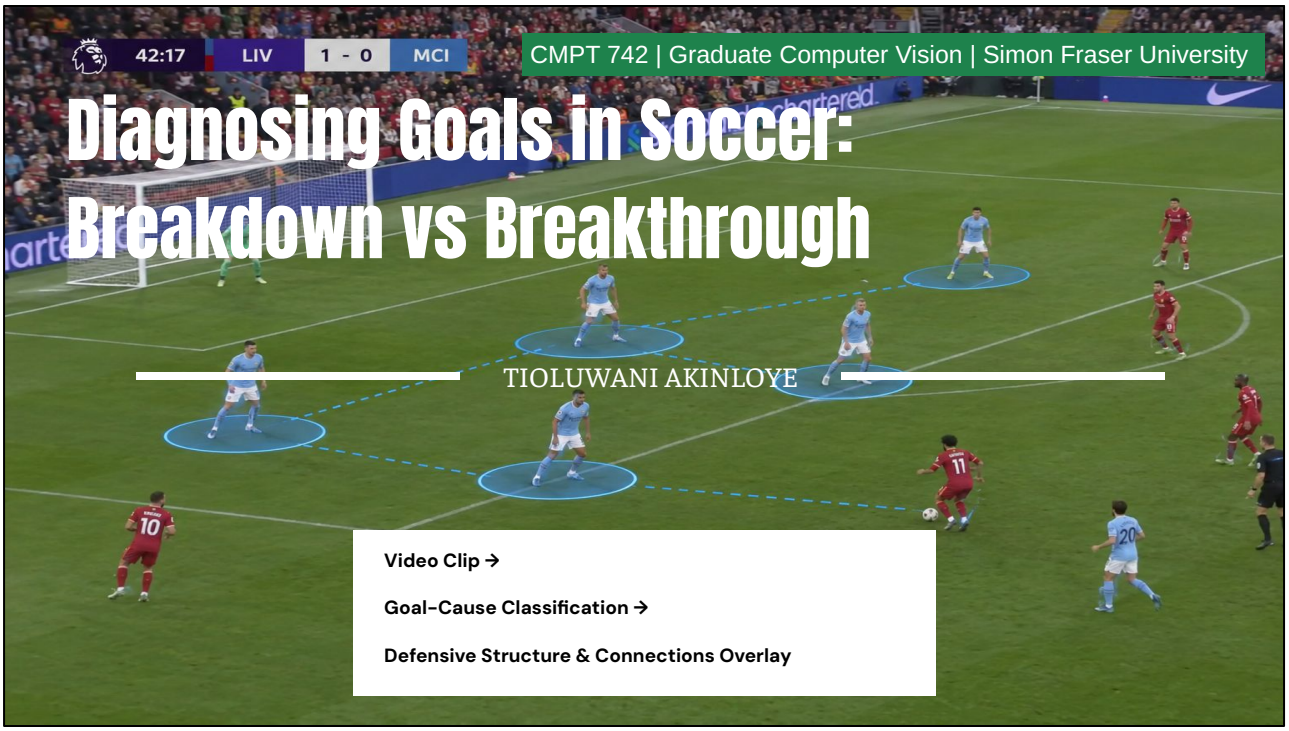


0:30

Beat 1: Introduction (~30 sec)
“My name is Tioluwani Akinloye, and today I’ll be giving a milestone presentation on my project, Diagnosing Goals in Soccer: Breakdown vs. Breakthrough.

This project focuses on the interpretable analysis of conceded goals through defensive line and pressure analysis.

Given a clip of a goal, my system aims to determine whether the goal was the result of the defense breaking down or whether the attack simply broke through, and then displaying the evidence over top the video.”



2:00

Beat 1: Personal hook (~20 sec)
I've played soccer since I was a kid, defense and wide midfield.
"I've played soccer since I was a kid, my main positions being either as a defender or a defensive midfielder. So I've spent a lot of time on the side of the game where you're responsible when things go wrong."

Beat 2: The moment (~30 sec)
You concede, and the first question is always "how did that happen?"
"Every player knows the moment when a goal is scored against you, everyone is asking: how did that happen? Did we mess up or was there nothing that could be done?"

Beat 3: How it's answered today (~35 sec)
Someone replays the clip and judges by eye.
"Today that question is mostly answered by someone replaying the clip and judging by eye. A coach tells the team they're out of shape, or a teammate counters there is nothing that could have been done. That's **slow, subjective, and hard to scale,** especially across a whole season."

Beat 4: Why it's a real problem (~30 sec)
Same goal, two different verdicts.
"It is also subjective in multiple ways. Two people can watch the same goal and disagree on whether the defense was at fault. If you want to improve a defense, you

need to know whether the problem was the execution or the opposition."

Beat 5: What you want to do (~25 sec)
Take one clip of a goal and explain it from the geometry.
"So my goal is simple to say: take a clip of the build-up to a goal and use the **positions of the defenders** to say whether or not the defense broke down or did their job, and then to show the evidence on the video so anyone can check it."

Beat 6: Transition (~10 sec)
"So why does it matter if we can answer that better?
"So if we are able to answer that question objectively, what does that actually change? That's why I think this problem matters."



1:30

Beat 1: Training (~25 sec)
Shape problems and opposition problems need different fixes.
"The first reason is training. If a goal came from a **broken shape**, the fix is on the pitch: spacing, line height, shifting together. If it came from a great attack, drilling the defense on the same tactic won't help much. Knowing which one it was tells a coach where to spend their time, whether changing strategies or fixing the current one"

Beat 2: Scale (~25 sec)
One goal by eye is fine, a whole season isn't.
"The second is scale. Reviewing one goal by eye is fine, but across a season or a league nobody has time to do that for every conceded goal. An automated, explainable diagnosis could **flag patterns**, like a team that keeps getting caught with the same gap, that one would be hard pressed to spot otherwise."

Beat 3: Product (~20 sec)
Potential post-match tool for analysts and coaches.
"As a product, this could be expanded upon to become a **post-match tool**: upload games, auto detect goal and input the last 15secs of a goal into this system, get the label and the faults highlighted on the video."

Beat 4: Research (~15 sec)
Single-goal, video-only diagnosis is a gap.
"As research, I wasn't able to find work that diagnoses **a single goal from video**, or

even address similar concerns, so even a working proof of concept would be a useful contribution. Especially since in the few papers I read, even defensive analysis was referred to as understudied and the paper urge further work building on it's conclusions"

Beat 5: Transition (~10 sec)
"So why isn't this already solved?"
"So if this is useful, the obvious question is why isn't it already done. It turns out there are a few things that make it harder than it looks."



2:00

Beat 1: Defense is overlooked (~20 sec)
Attack dominates the analysis, while defense is often overshadowed.
"The saying goes, *'The best defense is a good offense.'* And that mindset is reflected in how we consume soccer. Some of the most popular players are strikers because they score the goals and create these big moments that get headlines. Goals are easy to see, easy to count, and easy to celebrate. But preventing a goal is just as important as scoring one, even if it's harder to notice. Because goals aren't just scored because of great attacks; they're also conceded because of defensive breakdowns. If we want to understand why goals happen, we need to give defense the same analytical attention as attack.

Beat 2: No consistent measure (~20 sec)
Defending is subjective, so there's no consistent way to measure it.
"The first difficulty is that defending is largely subjective. Ask two people whether a defense was out of shape and you can get two different answers. There's no agreed way to measure it. So I can't just train using a label, one needs to be defined.

Beat 3: Defense is a collective (~35 sec)
Existing methods credit one or two players, or boil defensive performance down to statistics.
"Second, defense is a collective shape, and that's hard to capture. Alot of the existing approaches I've seen, credit the players closest to the action. Others average many

sequences into statistics or predictions. Both are useful, but neither gives you a discrete, per-goal explanation you can point at on screen and say, ‘There, that’s the gap.’”

Beat 4: Scale (~20 sec)

By eye across a season doesn’t work.

“And the manual alternative doesn’t scale. Reviewing clips by eye at the end of a season is **tardy, and variable from reviewer to reviewer**, so patterns across many goals are easy to miss.”

Beat 5: Camera (~25 sec)

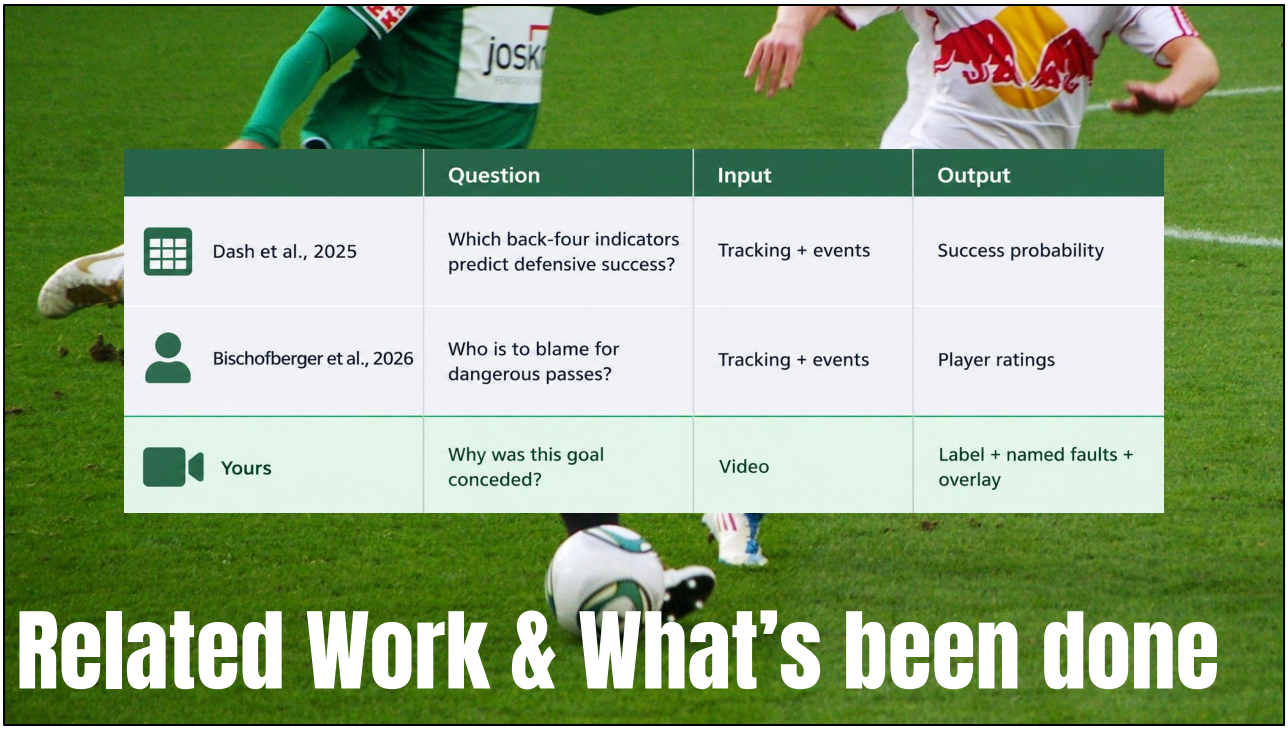
Cameras pan and zoom, so pitch coordinates are hard.

“Third, from video, I need to know where players are on the pitch. But match footage usually comes from cameras that **pan and zoom** and even the best tactical videos I’ve found still pan, given how big soccer fields are, so the mapping from image to pitch changes every frame, and need to be re-estimated constantly.”

Beat 6: Transition (~10 sec)

“So what has been done?”

“So with those obstacles let’s see what’s been done so far, and where they stops.”



1:30

Beat 1: Setup (~20 sec)
Recent defensive analysis mostly starts from tracking data.
"Recent work on evaluating defense mostly starts from **tracking data**, meaning the positions of all 22 players and the ball in every frame. I found 2 papers that were closest to my project."

Beat 2: Dash et al. (~30 sec)
Back-four indicators that predict defensive success.
"Prediction-based evaluation of back-four defense with spatial control in soccer (Paper 1). defines indicators for the **back four players** in the moments following a loss of possession, such as line height relative to the ball, space control, and stretch. They analyze about 2,400 defensive transitions from two LaLiga teams, and use statistics and machine learning to predict whether the defense succeeded (prevented a shot attempt). Relative line height came out strongest, which supports the kind of line metrics I'm using."

Beat 3: Bischofberger et al. (~25 sec)
Blame for individual defenders, from pass value.
"Blame is easier than praise: Measuring off-ball defensive performance in football (Paper 2). goes player by player. They value each pass and assign **blame or credit** to defenders within a fixed zone around the pass, then validate against market values and game ratings. It's a season-level rating of players, not an explanation of one

goal.”

Beat 4: Video side (~15 sec)

Video pipelines get to positions and stop.

“On the video side, work like SoccerNet’s game state reconstruction turns broadcast footage into player positions on a minimap, but as far as I found it **stops at positions and classification data,** but doesn’t produce any meaningful tactical inferences.”

Beat 5: Gap (~10 sec)

I haven’t found single-goal, video-based diagnosis.

“I haven’t found work that **diagnoses a goal given a video**, or provides named faults shown on top of the footage. That’s the gap I’m targeting.”

Beat 6: Transition (~5 sec)

“So what’s the obvious baseline?”

“So before my approach, what’s the simplest thing you could do?”



1:30

Beat 1: Setup (~10 sec)
What's the simplest thing you could do?
"So what's the obvious way to attack this? There are a few simple approaches, and they help to highlight where mine adds something unique."

Beat 2: Single-metric rule (~30 sec)
One number, one threshold.
"The simplest is a **single-metric rule**. For example, flag a breakdown whenever fewer than some number of defenders are goal-side of the ball. It's easy and it's explainable, but it's **one number**. It can't tell you a line had a gap or that one defender lagged behind, and it misses faults that don't show up in that one count. In my design, it's basically a single row of my own fault table."

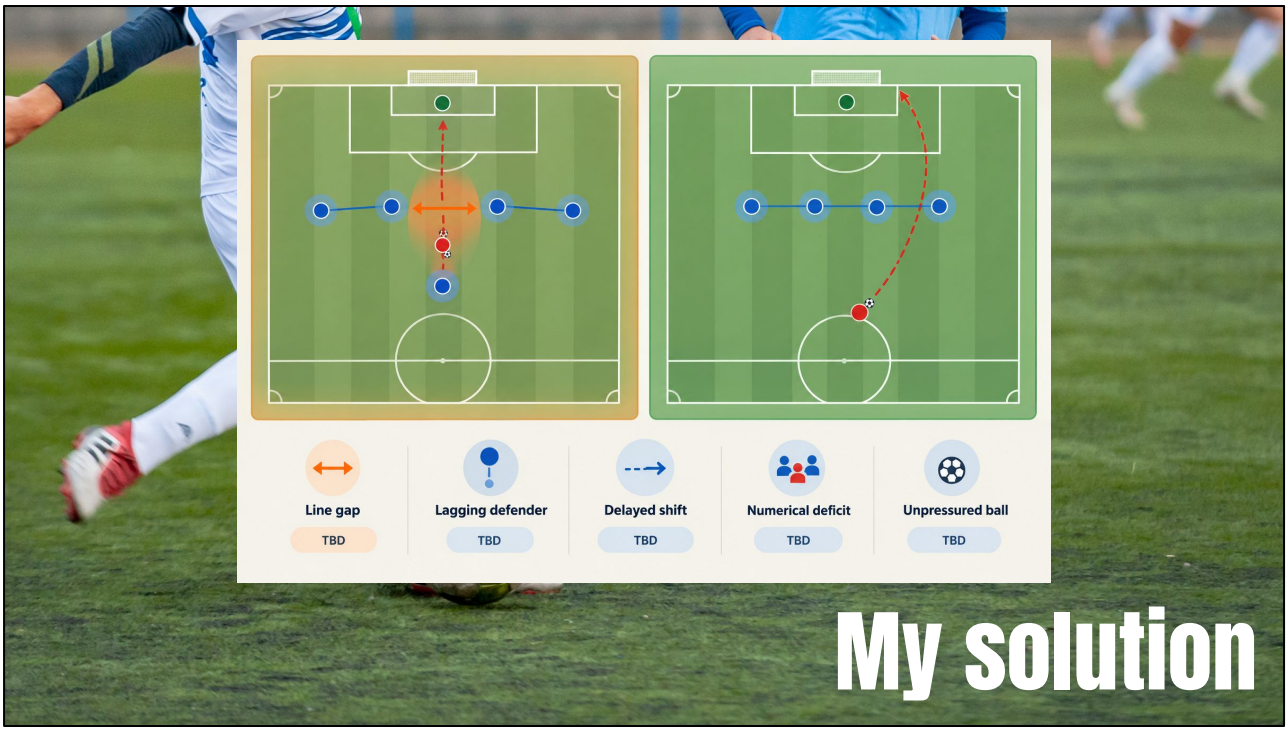
Beat 3: Off-the-shelf pipeline (~20 sec)
Positions, but no interpretation.
"The second is an **off-the-shelf, regular degular detection and tracking pipeline**. It gives you player positions and identifies them, but it stops there. Nothing in it says what the shape means. That's where my work starts."

Beat 4: Trained classifier (~20 sec)
Needs labeled goals that don't exist.
"A third idea is to **train a classifier**. But that needs a large set of labeled goals, and there's **no ground truth** for this. Even if it worked, it would give a score and not a

reason.”

Beat 5: Why mine (~15 sec)
Several features, set definitions, named faults, shown on video.
“My approach combines **several features under fixed definitions**, so each label comes with the **named fault** that triggered it, shown on the video. I’d say it’s better on interpretability but I don’t think I’m able to claim better accuracy yet.”

Beat 6: Transition (~5 sec)
“So here’s how it works.”
“But let me walk through how the pipeline actually works.”



1:30

Beat 1: Setup (~10 sec)
I need to be precise about the labels.
"To judge a goal fairly, I first need to be exact about what I mean by the two labels."

Beat 2: Breakdown (~20 sec)
At least one structural fault in the window before the shot.
"A **breakdown goal** is one where at least one **formational or positional fault** occurs in a short window before the shot. By fault I mean things like a broken line, a big gap, or a defender out of position."

Beat 3: Breakthrough (~15 sec)
Scored despite a defense that held its shape.
"A **breakthrough goal** is the opposite: the goal is scored **even though the defense kept its shape**. In practice that means no fault was detected under my rules."

Beat 4: Operational, not causal (~25 sec)
Measured geometry, not the true cause.
"These are **operational definitions**. They describe measured geometry, not the true cause of the goal or the quality of either team. I deliberately leave out one-on-one defending and goalkeeping, and that follows the feedback you gave on avoiding subjective definitions."

Beat 5: Fault list (~30 sec)
Five geometric faults, two spoken through.

“There are **five faults**, all purely geometric. A **line gap** is when the space between neighboring defenders gets too wide. A **lagging defender** is too far ahead of or behind the defensive line. A **delayed shift** is when the line fails to move across with the ball quickly enough. A **numerical deficit** means too few defenders are goal-side of the ball. And an **unpressured ball** is when no defender gets close enough to challenge the ball carrier for too long.”

Beat 6: Thresholds (~10 sec)

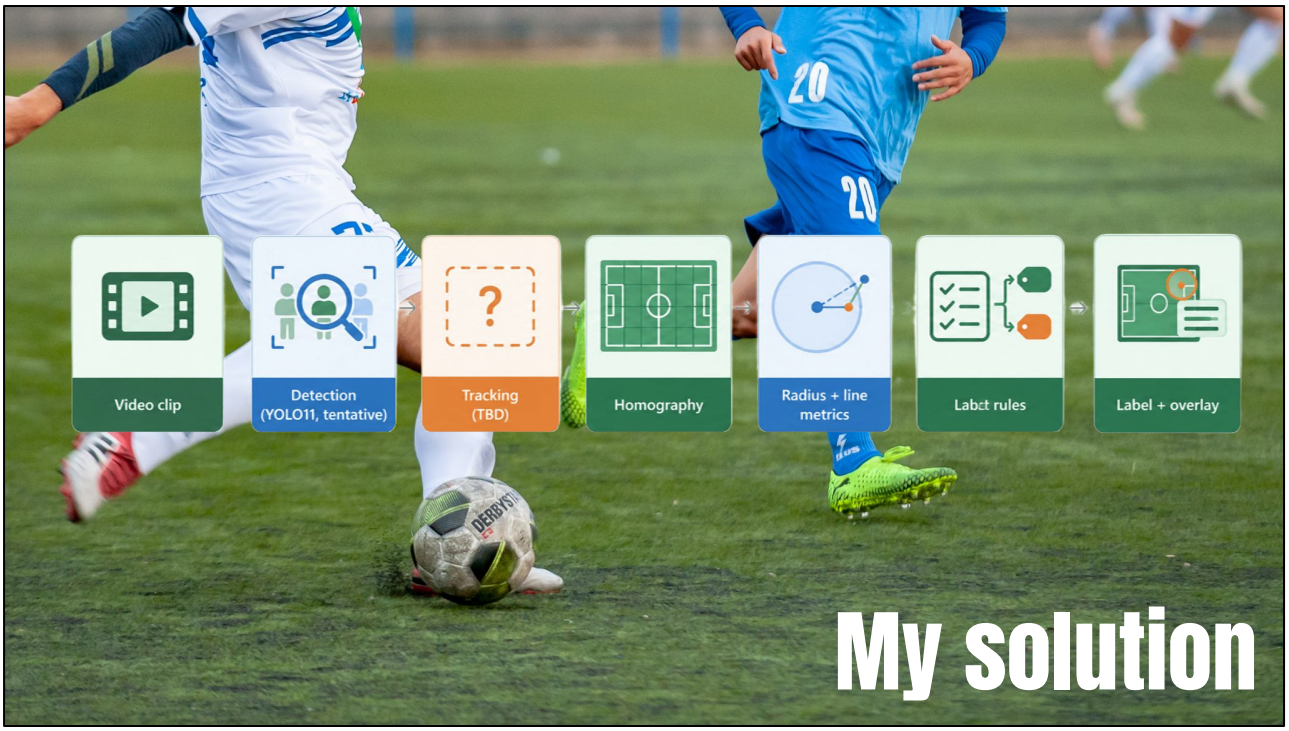
Values aren’t set yet.

“The threshold values aren’t set yet. They are **tunable parameters** that I plan to choose later and then tune through rustic sensitivity analysis. (changing one at a time and seeing the output and then adjusting)”

Beat 7: Transition (~5 sec)

“Here’s how the system finds them.”

“So how does the system find these faults from raw video? Let me walk through that pipeline.”



1:30

Beat 1: Overview (~10 sec)
Video in, label and overlay out.
"The pipeline takes a **video clip in** and gives a **label and an overlay out**, in a few stages."

Beat 2: Vision stages (~25 sec)
Detection, tracking (TBD), homography.
"First I **detect** the players and the ball. I'm planning to use YOLO, though I'm playing around with specific models at the moment. Then I **track** them across frames, the method for this is still tbd. Then a **homography** maps the image onto pitch coordinates, so everything is in real positions."

Beat 3: Metrics (~25 sec)
Radius around each defender, line between defenders.
"From those positions I compute two things. A **defensive radius** around each defender, which starts as a fixed distance, with speed-based versions later if time allows. And the **line connecting the defenders**, from which I get gaps, line height, and compactness. Related papers use radii of roughly 3 to 5 meters, so that's where I'll start."

Beat 4: Rules and output (~20 sec)
Fault rules over the window, then label plus overlay.
"The **fault rules** then run over the final few seconds before the shot. The output is

the label, the named faults that triggered it, and the radii and lines overlaid on the video."

Beat 5: Soccer line (~10 sec)
Formalizing "holding the line."
"As a defender, you hear 'hold the line' and 'stay connected' all the time. This is my attempt to **put numbers on that**."

Beat 6: Why mine (~10 sec)
Named faults, shown on video.
"What this adds over the simple baselines is that every label comes with a **named reason you can see**."

Beat 7: Transition (~5 sec)
"So where am I now?"
"As for that's the design. Here's what I have working so far."



1:30

Beat 1: Framing (~10 sec)
Setup phase is done, vision stage is in progress.
"So where am I? About three weeks in, the **setup phase is done** and I'm in the middle of the **vision stage**."

Beat 2: Done (~30 sec)
Lit review, repo, environment, sprint plan.
"On the done side: the **literature review** is finished, the **repo and environment** are set up, and I have a clear plan. I modelled it like a **sprint**, with user stories, deadlines, and pull requests, so each piece of the pipeline has an branch and a date."

Beat 3: Data (~25 sec)
Free, consistent bird's-eye source found.
"I also found a **free and consistent source** of bird's-eye or tactical-view footage. That matters because a consistent view means the camera doesn't change mid-clip, which is the biggest risk for the pitch-coordinate step."

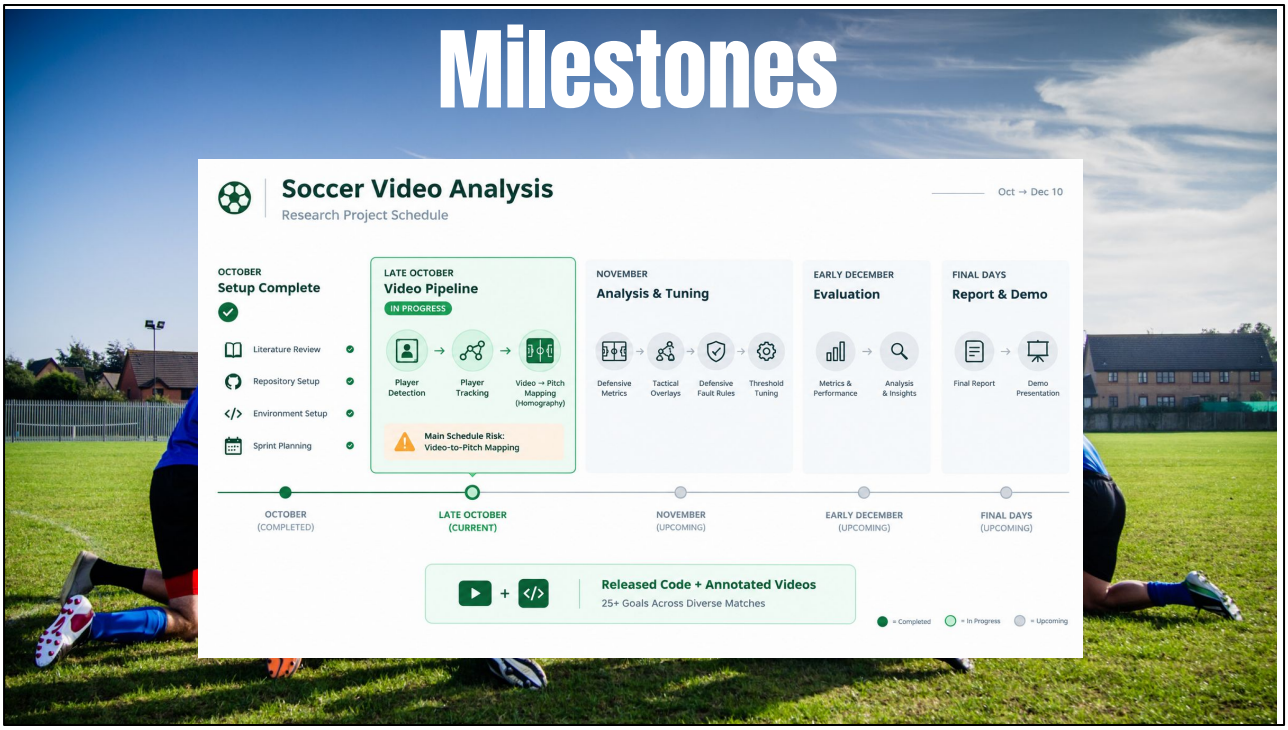
Beat 4: Detection, in progress (~35 sec)
Comparing YOLOv8 and YOLO11 on speed vs. accuracy.
"The part I'm working on now is **detection**. YOLO is already set up, and I'm comparing **YOLOv8 against YOLO11** for speed versus accuracy. Both are pretrained on general objects, so the next question is **adapting them to football footage**. I'm looking at two options: using a **free soccer dataset** with similar camera angles and

dimensions, or **building my own labeled set**[, with Roboflow to help label and split train, validation, and test data]."

Beat 5: Not yet (~20 sec)
Tracking, homography, overlays come next.
"What's **not done yet**: tracking, which is still undecided, the homography to pitch coordinates, and the overlays. I don't have results to show on those yet, but I have a vision."

Beat 6: Risk (~10 sec)
Video stage is the riskiest part.
"The video stage is the **riskiest part** of the project, so I'm hoping to get it done first."

Beat 7: Transition (~5 sec)
"Here's the timeline."
"Now, here's how the rest of the semester is laid out."



1:00

Beat 1: Overview (~10 sec)
October to about December 10.
"The current plan runs from **October to about December 10**, so a little under two months and it's laid out as a sprint with deadlines for each piece. (I'm not sure if this is a one semester scale project or a two semester scale project, initially I thought it may be one semester, b I put my proposal through different LLMs and the consensus seems to be ambivalent, but I plan to consult you about this.),"

Beat 2: Where we are (~15 sec)
Setup done, now in detection and tracking.
"The setup phase is **done**. We're here, at the end of October, in the **video stage**: detection, tracking, and the pitch homography."

Beat 3: November (~15 sec)
Metrics, overlays, fault rules, tuning.
"November is for the **defensive metrics and overlays**, then turning them into the **fault rules** and tuning the thresholds."

Beat 4: December (~10 sec)
Evaluation early, report and demo last.
"Early December is **evaluation and analysis**. The last days are for the **report and demo**, since only about ten days remain by then."

Beat 5: Deliverable (~10 sec)

Code plus annotated videos for 25+ goals.
"The deliverable is **released code and annotated videos for at least 25 goals**, across a diverse set of matches."

Beat 6: Risk (~10 sec)
The video-to-pitch stage is the schedule risk.
"The main schedule risk is the **video-to-pitch stage**, which is why it's first."

Beat 7: Transition (~5 sec)
"Let me wrap up."
"So, to wrap up."



0:30

Beat 1: Callback (~10 sec)
“How did that happen?” is hard to answer by eye.
“I started with the question every player asks after a goal against: *how did that happen?* Today that’s answered by eye, which is slow and subjective.”

Beat 2: Problem and approach (~15 sec)
Video to pitch coordinates, geometric fault rules, named faults on the video.
“My approach turns **video into pitch positions**, applies **geometric fault rules**, and labels the goal as a breakdown or a breakthrough, with the **named fault shown on the footage**.”

Beat 3: Status and next (~10 sec)
Setup done, vision stage in progress, metrics and rules next.
“So far the **setup is done** and the **vision stage is in progress**. Next come the metrics and the fault rules. If it works, it’s a proof of concept with **potential** for coaches and for research.”

Beat 4: Ask (~10 sec)
Feedback on the fault definitions and thresholds.
“One thing I’d really value is **feedback on the fault definitions, thresholds, and feasibility of timeline** since those are the part with no standardized truths.”

Beat 5: Close (~5 sec)
Thank you, questions.

“Thank you, and I’m happy to take further questions.”


Conclusion
THANK YOU!
Questions?